

Digital Marketing Tool

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Abstract—This project presents a Digital Marketing Funnel Management System implemented in the C programming language using a Binary Search Tree (BST). The system is designed to manage different stages of a marketing funnel by allowing users to perform operations such as insertion, searching, updating, deletion, and display of stages. Each stage is stored as a node containing a stage name and description, organized efficiently based on lexicographical order. The project highlights key concepts such as dynamic memory allocation, recursion, and hierarchical data representation while ensuring efficient data handling.

Key words- Binary Search Tree (BST), Digital Marketing Funnel, Data Structures,

Introduction

A digital marketing funnel represents the journey of a customer from awareness to conversion. Managing these stages effectively is essential for understanding user behavior and improving marketing strategies. This project focuses on building a simple yet efficient system to manage funnel stages using a Binary Search Tree. By using this data structure, the system ensures organized storage and faster operations compared to linear approaches. The implementation also helps in understanding fundamental programming concepts in C, especially related to memory and pointers.

Objectives

The primary objective of this project is to implement a Binary Search Tree to manage digital marketing funnel stages efficiently. It aims to support basic operations such as creating, inserting, searching, updating, and deleting nodes while maintaining a structured hierarchy. Another goal is to demonstrate the use of dynamic memory allocation and recursion in solving real-world problems. Additionally, the project helps in visualizing hierarchical relationships among different funnel stages.

Scope of the Project

The scope of this project is limited to a console-based application that simulates a digital marketing funnel. It is mainly intended for educational purposes to understand data structures and algorithms. However, the system can be extended further by integrating graphical interfaces or database connectivity for real-world applications. It provides

a foundation for more advanced systems involving customer analytics and marketing automation.

LITERATURE REVIEW

The project is based on the concept of Binary Search Trees, which are widely used for efficient searching, insertion, and deletion operations. BST organizes data in a hierarchical manner, reducing time complexity compared to linear structures. The implementation also relies on dynamic memory allocation using functions like malloc and free, which allow flexible memory usage during runtime. Recursive techniques are used for tree traversal and manipulation, which simplifies complex operations. These concepts are fundamental in computer science and widely applied in various software systems.

Methodology

The system begins by creating a root node representing the initial stage of the funnel. Additional stages are inserted into the tree based on alphabetical order to maintain BST properties. The program allows users to perform various operations such as searching for a stage, updating its description, deleting a stage, and displaying the entire funnel structure. Recursive algorithms are used for traversal and modification of the tree. The methodology ensures that all operations are performed efficiently while maintaining the integrity of the data structure.

Technology Used

The project is implemented using the C programming language, which provides low-level memory control and efficient execution. Key technologies and concepts used include data structures (Binary Search Tree), pointers for memory referencing, dynamic memory allocation for flexible storage, and recursion for implementing tree operations. Standard libraries such as stdio.h, stdlib.h, and string.h are used for input/output operations, memory management, and string handling.

Architecture / System Design

The system is designed using a tree-based architecture where each node represents a funnel stage. Each node contains a stage name, a description, and pointers to left and right child nodes. The left subtree contains nodes with smaller stage names, while the right subtree contains larger ones, maintaining the BST property. The design ensures efficient organization and retrieval of data, making it suitable for hierarchical representations like marketing funnels.

Implementation Details

The implementation includes several functions to handle different operations on the tree. The `createNode` function initializes a new node with dynamic memory allocation. The `insertNode` function places nodes in the correct position based on BST rules. The `searchNode` function locates a specific stage efficiently. The `updateNode` function modifies the description of an existing stage, while the `deleteNode` function removes a node and restructures the tree accordingly. Additional functions such as `displayTree`, `countNodes`, `showLeafNodes`, and `treeDepth` provide insights into the structure and properties of the tree. Memory management is handled using the `freeTree` function to prevent memory leaks.

Results / Output

The system successfully allows users to manage funnel stages dynamically through a menu-driven interface. Users can add new stages, search for existing ones, update descriptions, and delete stages as needed. The tree structure is displayed in a hierarchical format, providing a clear visualization of relationships between stages. The program also calculates the total number of stages, identifies leaf nodes, and determines the depth of the funnel, demonstrating the effectiveness of the

BST implementation. Scalable, and serves as a strong foundation for more advanced applications. Overall, it highlights the practical use of data structures in solving real-world problems.

Future Work

The project can be enhanced by adding a graphical user interface to improve user interaction and usability. Integration with databases can allow persistent storage of funnel data. Implementing self-balancing trees such as AVL or Red-Black Trees can improve performance for larger datasets. Additional features like analytics, reporting, and visualization tools can also be included to make the system more practical for real-world digital marketing applications.

Footnotes

The concepts used in this project are derived from standard data structures and C programming resources. Reference materials include textbooks on data structures covering Binary Search Trees and recursion, as well as documentation for standard C libraries such as `stdio.h`, `stdlib.h`, and `string.h`. Online programming resources and tutorials were also used to understand implementation techniques and best practices.

REFERENCES

- [1] GeeksforGeeks, "Dynamic Memory Allocation in C". Available at: <https://www.geeksforgeeks.org/dynamic-memory-allocation-in-c-using-malloc-calloc-free-and-realloc/>. B. Rieder, *Engines of Order: A M*
- [2] Infocodify, "Data Structures in C Programming". Available at: <https://www.infocodify.com/c-programming/data-structures>